Binomial simulations

Entries are mean (standard error) across replicates.

Model 1

uniLasso, AR(1) rho = 0.8, class mean shift in x1−x20; n = 100, p = 300, 30 nonzero

	sse	rpe_class	rpe_kl	support	sign	tpr	fpr
suni2D_zeta_tau	22.4293 (0.1007)	3.4778 (0.2719)	2.9408 (0.1225)	28.2350 (0.3667)	0.9041 (0.0014)	0.4910 (0.0049)	0.0500 (0.0013)
suni2D_lambda_alpha	22.6643 (0.1436)	3.4815 (0.2712)	2.9836 (0.1233)	28.6100 (0.3862)	0.9031 (0.0014)	0.4925 (0.0050)	0.0512 (0.0014)
suni2D_sum_rate	22.6656 (0.1415)	3.4746 (0.2708)	2.9851 (0.1233)	28.6350 (0.3837)	0.9030 (0.0014)	0.4923 (0.0050)	0.0514 (0.0014)
uniLasso	21.4873 (0.1547)	2.8018 (0.1884)	2.3891 (0.1012)	15.8650 (0.2257)	0.9406 (0.0006)	0.4675 (0.0055)	0.0068 (0.0005)
lasso	22.4351 (0.1026)	3.4095 (0.2344)	2.9527 (0.1220)	28.5200 (0.3806)	0.9034 (0.0014)	0.4922 (0.0050)	0.0509 (0.0014)
adalasso_uni	20.9708 (0.1604)	2.7638 (0.2108)	2.3353 (0.1131)	17.7900 (0.2038)	0.9403 (0.0007)	0.4983 (0.0047)	0.0105 (0.0005)
adalasso_lasso	28.5849 (0.2602)	4.0390 (0.3848)	3.5543 (0.1787)	12.4700 (0.1582)	0.9179 (0.0006)	0.2975 (0.0035)	0.0131 (0.0005)

Model 2

independent; n = 200, p = 1000, 10 nonzero (+/−1)

	sse	rpe_class	rpe_kl	support	sign	tpr	fpr
suni2D_zeta_tau	5.7882 (0.0718)	0.7212 (0.0169)	0.5584 (0.0084)	54.2300 (1.3091)	0.9547 (0.0013)	0.9485 (0.0058)	0.0452 (0.0013)
suni2D_lambda_alpha	5.7880 (0.0714)	0.7216 (0.0170)	0.5581 (0.0083)	54.2400 (1.2984)	0.9547 (0.0013)	0.9490 (0.0057)	0.0452 (0.0013)
suni2D_sum_rate	5.7881 (0.0714)	0.7218 (0.0169)	0.5582 (0.0083)	54.3350 (1.2982)	0.9546 (0.0013)	0.9490 (0.0057)	0.0453 (0.0013)
uniLasso	5.0193 (0.0853)	0.7127 (0.0203)	0.5226 (0.0101)	28.5450 (0.6125)	0.9791 (0.0005)	0.8830 (0.0089)	0.0199 (0.0006)
lasso	5.7856 (0.0717)	0.7203 (0.0169)	0.5581 (0.0083)	54.3150 (1.3099)	0.9547 (0.0013)	0.9485 (0.0058)	0.0453 (0.0013)
adalasso_uni	4.8690 (0.0932)	0.7528 (0.0162)	0.7061 (0.0172)	60.3400 (0.7790)	0.9487 (0.0008)	0.9545 (0.0048)	0.0513 (0.0008)
adalasso_lasso	6.6828 (0.1753)	0.6930 (0.0176)	0.9320 (0.0256)	30.5750 (0.4991)	0.9779 (0.0005)	0.9220 (0.0070)	0.0216 (0.0005)

Model 3

AR(1) rho = 0.5; n = 200, p = 1000, 10 nonzero (+/−1)

	sse	rpe_class	rpe_kl	support	sign	tpr	fpr
suni2D_zeta_tau	9.5872 (0.0309)	1.0431 (0.0173)	0.4735 (0.0050)	13.1900 (1.1360)	0.9806 (0.0010)	0.1890 (0.0117)	0.0114 (0.0011)
suni2D_lambda_alpha	9.5873 (0.0308)	1.0409 (0.0173)	0.4735 (0.0050)	13.1400 (1.1355)	0.9806 (0.0010)	0.1885 (0.0117)	0.0114 (0.0011)
suni2D_sum_rate	9.5875 (0.0308)	1.0405 (0.0174)	0.4735 (0.0050)	13.1600 (1.1341)	0.9806 (0.0010)	0.1885 (0.0116)	0.0114 (0.0011)
uniLasso	9.3979 (0.0379)	1.0010 (0.0163)	0.4865 (0.0058)	11.7700 (0.5549)	0.9823 (0.0005)	0.2045 (0.0093)	0.0098 (0.0005)
lasso	9.5864 (0.0309)	1.0379 (0.0173)	0.4733 (0.0050)	13.1650 (1.1368)	0.9806 (0.0010)	0.1890 (0.0117)	0.0114 (0.0011)
adalasso_uni	10.0095 (0.0868)	0.9928 (0.0131)	0.8242 (0.0191)	59.2350 (1.1724)	0.9391 (0.0011)	0.4175 (0.0096)	0.0556 (0.0011)
adalasso_lasso	10.7017 (0.1898)	1.0370 (0.0156)	0.8112 (0.0300)	10.4950 (0.7776)	0.9831 (0.0006)	0.1775 (0.0108)	0.0088 (0.0007)

Binomial simulations

Entries are mean (standard error) across replicates.

Model 4

block, size 20, rho = 0.5; n = 200, p = 1000, 10 nonzero (+/−1)

	sse	rpe_class	rpe_kl	support	sign	tpr	fpr
suni2D_zeta_tau	8.3473 (0.1016)	1.0484 (0.0255)	0.5434 (0.0073)	30.4400 (1.6694)	0.9690 (0.0013)	0.4725 (0.0230)	0.0260 (0.0015)
suni2D_lambda_alpha	8.3517 (0.1013)	1.0466 (0.0254)	0.5433 (0.0073)	30.4450 (1.6675)	0.9690 (0.0013)	0.4715 (0.0229)	0.0260 (0.0015)
suni2D_sum_rate	8.3523 (0.1014)	1.0474 (0.0255)	0.5434 (0.0073)	30.4100 (1.6683)	0.9690 (0.0013)	0.4715 (0.0228)	0.0260 (0.0015)
uniLasso	8.8513 (0.0721)	1.1213 (0.0216)	0.5739 (0.0068)	13.1300 (0.6462)	0.9829 (0.0004)	0.3040 (0.0142)	0.0102 (0.0005)
lasso	8.3515 (0.1013)	1.0468 (0.0254)	0.5433 (0.0073)	30.4000 (1.6672)	0.9691 (0.0013)	0.4735 (0.0230)	0.0259 (0.0015)
adalasso_uni	7.3495 (0.1050)	0.9376 (0.0198)	0.7292 (0.0140)	54.6500 (0.9163)	0.9486 (0.0008)	0.6645 (0.0130)	0.0485 (0.0009)
adalasso_lasso	9.9196 (0.2066)	0.9832 (0.0259)	0.9588 (0.0292)	19.8850 (0.9679)	0.9791 (0.0006)	0.4480 (0.0219)	0.0156 (0.0008)

Model 5

imbalanced, AR(1) rho = 0.5, prevalence 0.2; n = 200, p = 1000, 10 nonzero (+/−1)

	sse	rpe_class	rpe_kl	support	sign	tpr	fpr
suni2D_zeta_tau	9.7903 (0.0226)	1.2141 (0.0156)	1.0001 (0.0144)	9.9550 (0.9706)	0.9824 (0.0008)	0.1165 (0.0093)	0.0089 (0.0009)
suni2D_lambda_alpha	9.7892 (0.0227)	1.2142 (0.0156)	1.0002 (0.0144)	9.9450 (0.9747)	0.9824 (0.0008)	0.1165 (0.0093)	0.0089 (0.0009)
suni2D_sum_rate	9.7900 (0.0227)	1.2141 (0.0156)	1.0002 (0.0144)	9.9650 (0.9710)	0.9824 (0.0008)	0.1165 (0.0093)	0.0089 (0.0009)
uniLasso	9.7327 (0.0376)	1.2021 (0.0155)	1.0747 (0.0176)	11.9700 (0.6559)	0.9810 (0.0006)	0.1505 (0.0086)	0.0106 (0.0006)
lasso	9.7892 (0.0226)	1.2141 (0.0156)	0.9995 (0.0143)	9.8850 (0.9645)	0.9824 (0.0008)	0.1165 (0.0093)	0.0088 (0.0009)
adalasso_uni	11.1613 (0.1341)	1.1468 (0.0145)	1.8825 (0.0491)	50.4850 (0.9617)	0.9458 (0.0009)	0.3160 (0.0090)	0.0478 (0.0009)
adalasso_lasso	11.7626 (0.2761)	1.1705 (0.0152)	1.6408 (0.0652)	7.5050 (0.6628)	0.9847 (0.0006)	0.1085 (0.0087)	0.0065 (0.0006)

Model 6

Breheny−Huang (2011), independent; n = 400, p = 1000, 4 nonzero (+/−1)

	sse	rpe_class	rpe_kl	support	sign	tpr	fpr
suni2D_zeta_tau	0.9562 (0.0155)	0.1030 (0.0047)	0.1066 (0.0021)	39.6550 (1.5692)	0.9643 (0.0016)	1.0000 (0.0000)	0.0358 (0.0016)
suni2D_lambda_alpha	0.9564 (0.0154)	0.1025 (0.0046)	0.1067 (0.0021)	39.8950 (1.5768)	0.9641 (0.0016)	1.0000 (0.0000)	0.0360 (0.0016)
suni2D_sum_rate	0.9571 (0.0155)	0.1028 (0.0047)	0.1068 (0.0021)	39.7600 (1.5835)	0.9642 (0.0016)	1.0000 (0.0000)	0.0359 (0.0016)
uniLasso	0.3980 (0.0115)	0.0638 (0.0030)	0.0520 (0.0015)	15.0800 (0.5254)	0.9889 (0.0005)	1.0000 (0.0000)	0.0111 (0.0005)
lasso	0.9565 (0.0155)	0.1036 (0.0047)	0.1068 (0.0021)	39.7700 (1.5852)	0.9642 (0.0016)	1.0000 (0.0000)	0.0359 (0.0016)
adalasso_uni	1.4293 (0.0669)	0.2065 (0.0070)	0.2165 (0.0089)	69.6750 (1.7841)	0.9343 (0.0018)	1.0000 (0.0000)	0.0659 (0.0018)
adalasso_lasso	1.0658 (0.0533)	0.1525 (0.0062)	0.1576 (0.0074)	20.8850 (0.7331)	0.9831 (0.0007)	1.0000 (0.0000)	0.0170 (0.0007)